

COURSE CODE: CSC 311

CREDIT (C): 2

**COURSE TITLE: AUTOMATA THEORY, COMPUTABILITY
AND FORMAL LANGUAGES**

INTRODUCTION TO THEORY OF COMPUTATION

In theoretical computer science, the theory of computation is the branch that deals with whether and how efficiently problems can be solved on a model of computation, using an algorithm. The field is divided into three major branches: automata theory, computability theory and computational complexity theory.

What is computational theories?

The theory of computation is a branch of computer science and mathematics combined that "deals with how efficiently problems can be solved on a model of computation, using an algorithm". It studies the general properties of computation which in turn, helps us increase the efficiency at which computers solve problems. Theory of computation is the scientific discipline concerned with the study of general properties of computation and studies the inherent possibilities and limitations of efficient computation that makes machines more intelligent and enables them to carry out intellectual processes.

Theory of Computation also known as Computation Automata Theory is a theoretical branch of Computer Science and Mathematics, which mainly deals with the logic of computation with respect to simple machines, referred to as automata. Automata enables scientists to understand how machines compute the functions and solve problems.

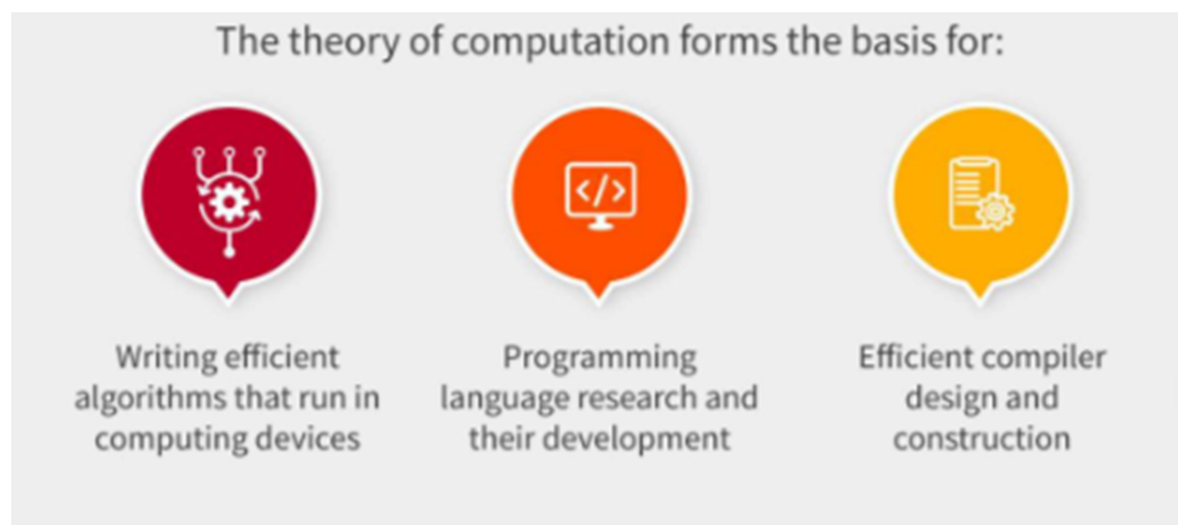
This theory enables scientists to understand how machines compute functions and solve problems. The main motivation behind developing Automata Theory was to develop methods to describe and analyze the dynamic behavior of discrete systems.

You can consider the theory of computation to be the creation of models of all sorts in the domain of computer science. Because of this, mathematics and logic are extensively used in it. In the 20th century, the theory of computation was separated from mathematics and became an independent academic discipline on its own.

In order to carry out a rigorous study of computation, computer scientists work with a mathematical abstraction of computers known as a model of computation. There are quite a few models of computation in use, but the Turing machine is the most examined one. Computer scientists study the Turing machine due to the fact that it is simple to formulate and can be analyzed and used to prove results. The Turing machine is also studied extensively because it represents what several computer scientists consider to be the most powerful possible "reasonable" model of computation. It could look like the potentially infinite memory capacity is an unrealizable attribute, but any decidable problem that can be solved through the use of a Turing machine will always need only a finite amount of memory.



Why is the theory of computation important?

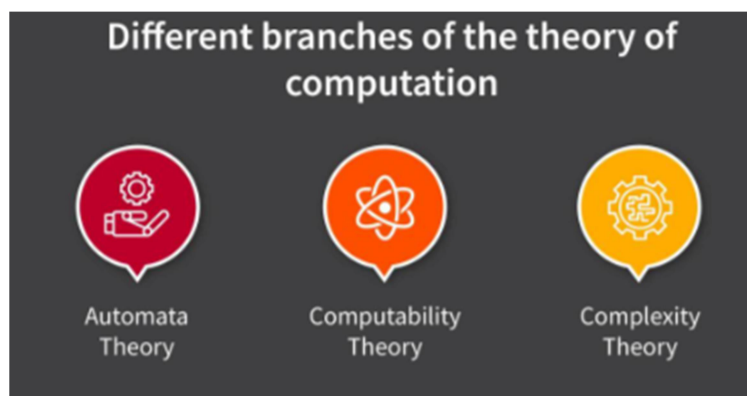


Real-world computers perform computations that by nature run like mathematical models to solve problems in systematic ways. The essence of the theory of computation is to help develop mathematical and logical models that run efficiently and to the point of halting. Since all machines that implement logic apply TOC, studying TOC gives learners an insight into computer hardware and software limitations.

The theory of computation forms the basis for:

- Writing efficient algorithms that run in computing devices.
- Programming language research and their development.
- Efficient compiler design and construction.

What are the different branches of the theory of computation?



The theory of computation is made up of 3 branches.

They are:

- **Automata Theory** - The study of abstract machines (or more appropriately, abstract 'mathematical' machines or systems) and the computational problems that can be solved using these machines. The abstract machines are known as automata. The automata theory was established by mathematicians in the 20th century. This branch of the theory of computation focuses on analyzing the behavior of machines and the manner in which they solve a problem. Automata is derived from the Greek word (Αυτόματα) which means that something is doing something on its own. The

most powerful model of automata is the Turing machine. Automata theory is rather closely related to formal language theory (a branch of mathematics that deals with describing languages as a set of operations over an alphabet) because the automata are quite frequently classified by the class of formal languages that they are able to recognize. An automaton could be a finite representation of a formal language that might be an infinite set. Automata are used as theoretical models for computing machines, and are used for proofs about computability.

- **Computability Theory** - This theory deals primarily with the question of the extent to which a problem is solvable on a computer. This is when the computer can address the problem, but it is unable to come up with the solution. The fact that the halting problem cannot be solved by a Turing machine is one of the most important results in computability theory, because it is an example of a concrete problem that is easy to formulate but, at the same time, impossible to solve through the use of a Turing machine. Quite a lot of computability theory builds on the halting problem result. Rice's theorem was another important step in computability theory. This theorem says that for all non-trivial properties of partial functions, it cannot be decided whether a Turing machine computes a partial function with that property.
- **Complexity Theory** - This theories considers not only whether a problem can be solved at all on a computer, but also how efficiently the problem can be solved. It discusses the efficiency with which a problem could be solved by considering two major aspects: time complexity and space complexity. These are measurements of the number of steps needed to analyze and solve the problem and thus determine the amount of memory space that is required in order to solve the problem. For the purpose of pre-

determining the amount of space and time that will be required, computer scientists linked these two factors to the amount of input that was received, as the time and space needed increase linearly as the amount of input increases.